

A Fast Group Sparse Phase Retrieval Algorithm for DOA Estimation Based on Polar-Coupled Array Architecture

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Abstract—This paper proposes a fast group-sparse phase retrieval algorithm based on a polar-coupled dual-channel array architecture (Polar-ToyBar). This algorithm addresses the challenge of Direction-of-Arrival (DOA) estimation in scenarios where only magnitude measurements are available, such as medical imaging through non-uniform tissues. By leveraging a dual-polarized array, we formulate the problem as a joint group-sparse phase retrieval task, where horizontal and vertical channels are coupled via a joint sparsity regularization term. The optimization problem is solved efficiently using a Majorization-Minimization (MM) framework combined with an accelerated proximal-gradient scheme. Simulation results demonstrate that the proposed method significantly enhances estimation accuracy and robustness compared to conventional methods like ToyBar, while effectively resolving spatial ambiguities.

Index Terms—DOA estimation, Phase retrieval, Polar-sensitive array, Group sparsity.

I. INTRODUCTION

Direction-of-Arrival (DOA) estimation is a fundamental problem in array signal processing, widely applied in radar, sonar, MIMO radar [1], and medical diagnosis systems [2]. Traditional subspace methods, such as MUSIC [3] and ESPRIT [4], rely on complex-valued array data containing both amplitude and phase information. However, in specific scenarios, such as low-cost automotive sensing, tracking signal sources inside the human body, or imaging through scattering media, phase information is often severely distorted or entirely lost. This loss typically stems from tissue attenuation, synchronization errors in distributed arrays, or hardware limitations, leaving only reliable magnitude measurements. This necessitates DOA estimation techniques based solely on magnitude data.

Existing phase retrieval algorithms can be broadly categorized into three types: alternating projection, convex relaxation, and non-convex optimization. Early methods like the Gerchberg-Saxton (GS) algorithm [5] use alternating projections but often suffer from stagnation at local minima. Convex relaxation methods, such as PhaseLift [6], lift the problem into a higher-dimensional space using Semidefinite Programming (SDP) to ensure global convergence, but they incur prohibitive computational costs. Recently, non-convex approaches like Wirtinger Flow (WF) [7] and the Greedy Sparse Phase Retrieval (GESPAR) algorithm [8] [9] have been developed to balance efficiency and accuracy. While GESPAR exploits sparsity, its computational load increases linearly with the number of snapshots. To address these limitations, the ToyBar algorithm [10]–[12] was introduced, utilizing a group-sparse model solved via Majorization-Minimization (MM) [13]. However, standard single-channel approaches struggle with inherent ambiguities, such as mirroring and spatial shifts, and often lack robustness in low Signal-to-Noise Ratio (SNR) environments. These limitations severely hinder the practical application of single-channel based methods in complex electromagnetic environments.

In this paper, we propose a Polar-Coupled ToyBar (Polar-ToyBar) algorithm. Specifically, we introduce a polar-sensitive array architecture designed to simultaneously capture orthogonal Horizontal (H) and Vertical (V) polarization components. By formulating a joint group-sparsity constraint, we effectively couple the recovery of H and V channels, ensuring they share the same spatial support. This dual-channel redundancy not only enhances robustness against noise but also utilizes a specific dual-subarray geometry to resolve phase retrieval ambiguities.

II. PROBLEM FORMULATION

To clearly formulate the problem, this section first establishes the signal model based on the polar-coupled dual-channel array architecture. Subsequently, we analyze the inherent ambiguities arising from magnitude-only measurements in DOA estimation and explain how the proposed array configuration mitigates these issues.

A. Signal Model

As shown in the figure, the structure consists of two uniform linear arrays, each with an inter-element spacing of d . The first sensor set is arranged horizontally, while the second is rotated by a known tilt angle $\tilde{\theta}$. For the effective aperture of each sub-array, incident directions are limited to $[-90^\circ, 90^\circ]$. Since both N -element arrays share a common sensor, the system totals $2N - 1$ effective sensors. The array positions are $\mathbf{r}_1 = \{0, d, 2d, \dots, (N-1)d\}$ and $\mathbf{r}_2 = \{\tilde{\theta}, \tilde{\theta} + d, \tilde{\theta} + 2d, \dots, \tilde{\theta} + (N-1)d\}$. The overall position set is $\mathbf{r} = \mathbf{r}_1 \cup \mathbf{r}_2$.

Each sensor in the dual-array is a dual-polarized antenna, capable of receiving signals from both the H and V orthogonal polarization channels separately. Assume that there are K far-field narrow band signals in the space, all incident from directions θ_k , $k = 1, \dots, K$, with the same wavelength λ . The noise-free received signal vector can be expressed as

$$\mathbf{x}_H = \mathbf{m} \mathbf{w}_H \quad (1)$$

$$\mathbf{x}_V = \mathbf{m} \mathbf{w}_V \quad (2)$$

where $\mathbf{m}$ is the extended steering vector of the dual-array, and the steering vectors of the two sub-arrays are respectively given by $\mathbf{m}_1(\theta) = [e^{-j\frac{2\pi}{\lambda} R_{1,1} \sin \theta_k}, \dots, e^{-j\frac{2\pi}{\lambda} R_{1,N} \sin \theta_k}]^T$ and $\mathbf{m}_2(\theta) = [e^{-j\frac{2\pi}{\lambda} R_{2,1} \sin \theta_k}, \dots, e^{-j\frac{2\pi}{\lambda} R_{2,N} \sin \theta_k}]^T$. At this point, $\mathbf{m} = [\mathbf{m}_1(\theta), \mathbf{m}_2(\theta)]^T$. For each sensor, we simultaneously observe two orthogonal polarization components: H (horizontal polarization) and V (vertical polarization).

For the k -th target, its polarization can be characterized by the amplitude ratio and phase difference, which is denoted as

$$w_{V,k} = \alpha_k e^{j\phi_k} w_{H,k} \quad (3)$$

The vectors $\mathbf{w}_H$ and $\mathbf{w}_V$ represent the source signal vectors for the H and V channels, where $\alpha_k = |w_{V,k}|/|w_{H,k}|$ and $\phi_k = \angle(E_V) - \angle(E_H)$. At the p -th snapshot, the array responses for the H/V channels are respectively given by

$$\mathbf{x}_H[p] = \mathbf{m} \mathbf{w}_H[p] \quad (4)$$

$$\mathbf{x}_V[p] = \mathbf{m} \mathbf{w}_V[p] \quad (5)$$

Based on the polarization coupling relationship, the source signals for the H/V channels can be expressed as $w_{V,k}[p] = \alpha_k e^{j\phi_k} w_{H,k}[p]$.

To utilize the sparse recovery framework, we discretize the interested DOA range $[-90^\circ, 90^\circ]$ into G grid points, forming an overcomplete steering matrix $\mathbf{M}(\theta)$.

$$\mathbf{M} = [\mathbf{m}(\theta_1), \dots, \mathbf{m}(\theta_G)] \quad (6)$$

where $\mathbf{M} \in \mathbb{C}^{(2N) \times G}$. Correspondingly, the signal vectors $\mathbf{w}_H[p]$ and $\mathbf{w}_V[p]$ are extended to sparse vectors of size $G \times 1$,

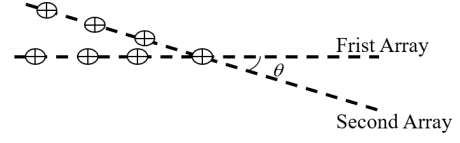


Fig. 1. The dual-subarray architecture with dual-channel reception.

denoted as $\mathbf{w}_H[p]$ and $\mathbf{w}_V[p]$. By collecting P snapshots, we obtain $G \times P$ -dimensional sparse signal matrices $\mathbf{W}_H$ and $\mathbf{W}_V$. These matrices exhibit joint row sparsity, meaning that they are non-zero on the same rows. The array reception model (with P snapshots) can be expressed in the sparse framework by stacking the matrices as follows

$$\mathbf{X}_H = \mathbf{M} \mathbf{W}_H \quad (7)$$

$$\mathbf{X}_V = \mathbf{M} \mathbf{W}_V \quad (8)$$

where the matrices $\mathbf{W}_H, \mathbf{W}_V \in \mathbb{C}^{G \times P}$ share the same sparse support (corresponding to the same set of DOAs) and satisfy the polarization relationship $\mathbf{W}_H = \mathbf{D} \mathbf{W}_V$, $\mathbf{D} = \text{diag}(\alpha_k e^{j\phi_k})$. In the system, we can only observe the magnitude information. Let $\mathbf{B}_H$ and $\mathbf{B}_V$ be the magnitude observation matrices, each of size $2N \times P$.

$$\mathbf{B}_H = |\mathbf{X}_H| + \mathbf{N}_H \quad (9)$$

$$\mathbf{B}_V = |\mathbf{X}_V| + \mathbf{N}_V \quad (10)$$

where $|\cdot|$ denotes the element-wise absolute value operation, $\mathbf{N}_H$ and $\mathbf{N}_V$ represent the additive white Gaussian noise.

B. Ambiguities

This study relies on magnitude-only measurements, leading to phase retrieval non-uniqueness. For a single array, this yields multiple ambiguous DOA solutions. Beyond standard global phase ambiguity, the non-coherent scenario introduces three more detrimental spatial ambiguities.

For any given angle θ , if the array has a geometrically center-symmetric structure, a mirroring ambiguity arises. The magnitudes generated by $[-\theta_1, \dots, -\theta_k]$ will be identical to those generated by $[\theta_1, \dots, \theta_k]$. The rotation angle of the dual-array structure breaks the mirror symmetry condition of the original array, thereby ensuring that angles with opposite signs generate different magnitudes.

For angle sets θ and θ' satisfying $\sin \theta'_k = \sin \theta_k + \Delta$, unmeasurable higher-order phase terms yield identical magnitude responses, causing global shift non-uniqueness in the DOA spectrum. However, the non-uniform sampling of the dual-array ensures magnitudes differ under such spectral shifts, effectively suppressing this ambiguity.

For a Uniform Linear Array with $d \geq \lambda/2$ spacing, when multiple DOAs are present simultaneously, certain combinations can lead to a reversal in the spatial frequency order, however, it still satisfies $|\mathbf{M}\mathbf{w}| = |\mathbf{M}\mathbf{w}'|$. The order of the DOAs and their relative positions may be rearranged, but the magnitudes remain consistent. This ambiguity can only be completely eliminated when the inter-element spacing is $d \leq \lambda/4$.

III. PROPOSED ALGORITHM

In this section, we recover the sparse matrices $\mathbf{W}_H$ and $\mathbf{W}_V$ from magnitude observations $\mathbf{B}_H$ and $\mathbf{B}_V$ by exploiting their shared row sparsity. Since the V channel snapshots are simply scaled versions of the H channel in the polarization model, both channels share a common angular support set. To enforce simultaneous activation at identical angles, a group sparsity norm is introduced

$$\mathcal{R}(\mathbf{W}_H, \mathbf{W}_V) = \sum_{g=1}^G \sqrt{\|\mathbf{W}_H(g, :)\|_2^2 + \|\mathbf{W}_V(g, :)\|_2^2} \quad (11)$$

To recover $\mathbf{W}_H$ and $\mathbf{W}_V$, we need to construct a joint fidelity term. We define the joint fidelity term as $\|\mathbf{B}_H - |\mathbf{M}\mathbf{W}_H|\|_F^2 + \|\mathbf{B}_V - |\mathbf{M}\mathbf{W}_V|\|_F^2$. Combining the above sparsity and fidelity terms, we formulate the problem as the following joint polarization non-coherent sparse optimization model

$$\min_{\mathbf{W}_H, \mathbf{W}_V} \|\mathbf{B}_H - |\mathbf{M}\mathbf{W}_H|\|_F^2 + \|\mathbf{B}_V - |\mathbf{M}\mathbf{W}_V|\|_F^2 + \gamma \mathcal{R}(\mathbf{W}_H, \mathbf{W}_V) \quad (12)$$

where $\|\cdot\|_F$ is the Frobenius norm, γ is the regularization parameter, and $\mathcal{R}$ is the joint group sparsity regularization term which encourages $\mathbf{W}_H$ and $\mathbf{W}_V$ to be sparse simultaneously on the same rows.

In the joint polarization non-coherent sparse optimization model above, the fidelity term is non-convex due to the modulus operators involved. To handle the non-convexity, we apply the Phase Retrieval via Majorization-Minimization (PRIME) algorithm [13].

At the q -th MM iteration, the non-convex fidelity term $\|\mathbf{B}_H - |\mathbf{M}\mathbf{W}_H|\|_F^2$ is majorized into a convex quadratic surrogate $\|\mathbf{C}_H^{(q)} - \mathbf{M}\mathbf{W}_H\|_F^2$. The auxiliary variable is $\mathbf{C}_H^{(q)} = \mathbf{B}_H \odot e^{j \arg(\mathbf{M}\mathbf{W}_H^{(q)})}$, where $\odot$ denotes the Hadamard product and $\mathbf{W}_H^{(q)}$ is from the previous iteration. Applying this to both channels separately, the $q+1$ -th iteration is replaced by solving the following convex problem

$$\min_{\mathbf{W}_H, \mathbf{W}_V} \|\mathbf{C}_H^{(q)} - \mathbf{M}\mathbf{W}_H\|_F^2 + \|\mathbf{C}_V^{(q)} - \mathbf{M}\mathbf{W}_V\|_F^2 + \gamma \mathcal{R}(\mathbf{W}_H, \mathbf{W}_V) \quad (13)$$

The original model is transformed into a convex problem of the following form $\min_{\mathbf{W}_H, \mathbf{W}_V} F(\mathbf{W}) + \mathcal{J}(\mathbf{W})$, where $F(\mathbf{W})$ is a smooth and differentiable fidelity term, and $\mathcal{J}(\mathbf{W})$ is a non-smooth but convex regularization term. Such problems can be efficiently solved using the proximal gradient method.

The problem is solved using the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) [14], which is based on the Nesterov accelerated proximal gradient method.

$$\min_{\mathbf{W}_H, \mathbf{W}_V} F(\mathbf{W}_H) + F(\mathbf{W}_V) + \mathcal{J}(\mathbf{W}_H, \mathbf{W}_V) \quad (14)$$

compute the gradient of $F(\mathbf{W}^{(q)})$ with respect to $\mathbf{W}_H^{(q)}$ and $\mathbf{W}_V^{(q)}$.

$$\nabla F(\mathbf{W}_H) = 2\mathbf{M}^H(\mathbf{C}_H^{(q)} - \mathbf{M}\mathbf{W}_H^{(q)}) \quad (15)$$

$$\nabla F(\mathbf{W}_V) = 2\mathbf{M}^H(\mathbf{C}_V^{(q)} - \mathbf{M}\mathbf{W}_V^{(q)}) \quad (16)$$

Perform gradient descent to obtain the intermediate matrices $\mathbf{U}_H$ and $\mathbf{U}_V$; $\mathbf{U}_H = \mathbf{W}_H^{(q)} - \eta \nabla F(\mathbf{W}_H)$ and $\mathbf{U}_V = \mathbf{W}_V^{(q)} - \eta \nabla F(\mathbf{W}_V)$. Then, in the $q+1$ -th iteration of FISTA, where λ is the step size, the proximal operator is defined as

$$\text{prox}_{\lambda G}(\mathbf{W}) = \arg\min_{\mathbf{Z}} \left(\frac{1}{2\lambda} \|\mathbf{Z} - \mathbf{W}\|_F^2 + G(\mathbf{Z}) \right) \quad (17)$$

The Lipschitz constant of the gradient of the quadratic term $\|\mathbf{C}_H^{(q)} - \mathbf{M}\mathbf{U}\|_F^2$ is $L_H = 2\lambda_{\max}(\mathbf{M}^H\mathbf{M})$, so the step size is chosen as $\eta < \frac{1}{L_H} = \frac{1}{2\lambda_{\max}(\mathbf{M}^H\mathbf{M})}$.

By replacing the first and second terms of the objective function with $F(\mathbf{W})$ and replacing the third term with $\mathcal{J}(\mathbf{W})$, the solution can be obtained by solving the following problem [15] [16]

$$\mathbf{W}^{q+1} = \arg\min_{\mathbf{Z}} \left\{ \frac{1}{2\lambda} \|\mathbf{Z} - (\mathbf{W}^q - \lambda \nabla F(\mathbf{W}^q))\|_F^2 + \gamma \|\mathbf{Z}\|_{2,1} \right\} \quad (18)$$

applying the row-wise approximation operator of the l_2 norm, its analytical solution is

$$\mathbf{W}^{q+1} = (\mathbf{W}^q - \lambda \nabla F(\mathbf{W}^q)) \cdot \max \left(1 - \frac{\gamma \eta}{\|\mathbf{W}^q - \lambda \nabla F(\mathbf{W}^q)\|_2}, 0 \right) \quad (19)$$

The positions of the non-zero rows in the reconstructed signal matrix $\mathbf{W}^q$ correspond to the DOAs of the incident signals.

By transforming the group sparsity phase retrieval problem into a convex algebraic form and solving it using the proximal gradient method, the problem can be further accelerated by applying Nesterov acceleration [14] [17]. This method is not directly applied to the previous $\mathbf{W}^{q+1}$, but is based on another point of $\mathbf{W}^{q+1}$ and $\mathbf{W}^q$, which is denoted as

$$\mathbf{D}^{q+1} = \mathbf{W}^{q+1} + \frac{\beta^q - 1}{\beta^{q+1}} \cdot (\mathbf{W}^{q+1} - \mathbf{W}^q) \quad (20)$$

where $\beta^{q+1} = \frac{1 + \sqrt{1 + 4(\beta^q)^2}}{2}$.

Upon convergence, the reconstructed sparse matrices $\hat{\mathbf{S}}_H, \hat{\mathbf{S}}_V \in \mathbb{C}^{G \times P}$ for both channels are obtained. Enforced by the joint group sparsity regularization, they share a common non-zero row support set corresponding to the true target locations.

To estimate DOA of the targets, the joint spatial power spectrum $P_{\text{spatial}}(\theta_g)$ at the g -th grid point is defined as the sum of the signal energies from both channels

$$P_{\text{spatial}}(\theta_g) = \|\hat{\mathbf{S}}_H(g, :)\|_2^2 + \|\hat{\mathbf{S}}_V(g, :)\|_2^2 \quad (21)$$

where $\hat{\mathbf{S}}_H(g, :)$ represents the g -th row vector of matrix $\hat{\mathbf{S}}_H$, and $\|\cdot\|_2$ denotes the l_2 norm. By searching for the local maxima of $P_{\text{spatial}}(\theta)$ over the grid $\Theta = \{\theta_1, \dots, \theta_G\}$, the estimated incident angles of the K targets can be determined, denoted as $\hat{\boldsymbol{\theta}} = \{\hat{\theta}_1, \dots, \hat{\theta}_K\}$.

After determining spatial locations, target polarization states are inverted via the amplitude differences between the recon-

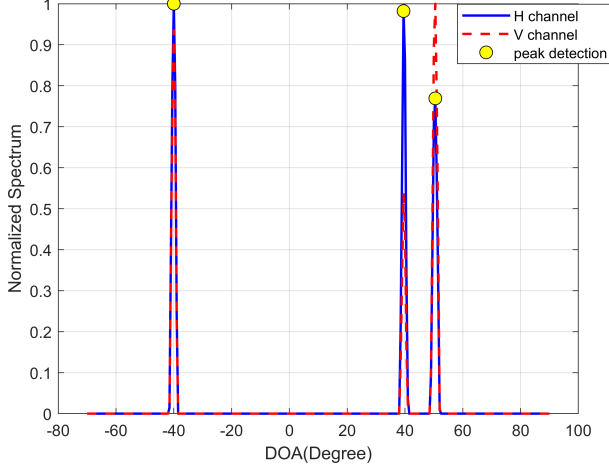


Fig. 2. Estimation results by the proposed Polar-ToyBar.

structed dual-channel signals. For the k -th estimated angle $\hat{\theta}_k$, the polarization amplitude ratio $\hat{\alpha}_k$ is

$$\hat{\alpha}_k = \frac{\|\hat{\mathbf{S}}_V(\hat{g}_k, :)\|_2}{\|\hat{\mathbf{S}}_H(\hat{g}_k, :)\|_2 + \epsilon} \quad (22)$$

where $\hat{g}_k$ is the grid index for $\hat{\theta}_k$, and ϵ (set to 10^{-6}) is a small constant preventing numerical instability when the horizontal signal is weak. The estimated $\hat{\alpha}_k$ directly characterizes target polarization properties. Subsequent simulations verify this recovery accuracy under non-coherent conditions by comparing $\hat{\alpha}_k$ with the true parameters α_{true} .

IV. SIMULATION RESULTS

This section provides simulation results to evaluate the proposed Polar-ToyBar. The dual-subarray architecture contains $N = 20$ omnidirectional sensors per subarray with $d = \lambda/4$ spacing to avoid spatial order ambiguity. A 0.5° scanning grid is used for sparse recovery. The regularization weight is fixed at $\gamma = 20$, and the step size λ is the reciprocal of the Lipschitz constant (twice the maximum eigenvalue of $\mathbf{A}^H \mathbf{A}$).

The first simulation presents the normalized spatial spectrum of the proposed algorithm at $SNR = 20\text{dB}$ with $P = 20$ snapshots. For $K = 3$ sources impinging from $\{-40^\circ, 40^\circ, 50^\circ\}$ with polarization ratios $\alpha_{\text{true}} = \{1.0, 0.8, 1.2\}$, Fig. 2 shows that both H (solid blue) and V (dashed red) channels yield sharp peaks accurately at the true DOAs. The method exhibits superior resolution, clearly distinguishing the closely spaced sources at 40° and 50° . Furthermore, the estimated polarization ratios $\hat{\alpha} = \{1.04, 0.81, 1.25\}$ match the ground truth with high precision.

TABLE I
Running times of different algorithms.

Snapshots	GESPAR(s)	ToyBar(s)	Polar-ToyBar(s)
P=20	118.07	29.52	58.79

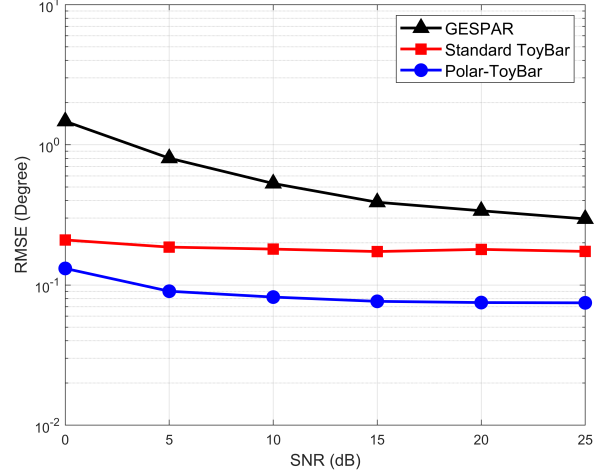


Fig. 3. RMSEs versus different SNR.

To quantify estimation accuracy, the second simulation compares the Root Mean Square Error (RMSE) of Polar-ToyBar against GESPAR and Standard-ToyBar. With $P = 100$ snapshots, Fig. 3 plots the RMSE versus SNR (0dB to 25dB) averaged over 100 Monte-Carlo trials. Polar-ToyBar consistently outperforms the others. Notably, for $SNR \leq 10\text{dB}$, it exhibits significantly lower error, validating the robustness gained by coupling dual-polarization information via the joint group sparsity term.

To evaluate computational complexity, TABLE I lists average running times on an AMD Ryzen 7 4800U CPU (1.80 GHz, 16 GB RAM). Polar-ToyBar demonstrates a significant speed advantage: at $P = 20$ snapshots, its 58.79 s runtime is approximately twice as fast as modified GESPAR (118.07 s). This efficiency stems from our joint group-sparsity penalty and accelerated proximal-gradient scheme bypassing GESPAR's iteration-heavy search. Furthermore, while processing dual-polarized channels adds overhead compared to standard ToyBar (29.52 s), the substantial improvements in accuracy, resolution, and robustness make this moderate trade-off completely worthwhile.

V. CONCLUSIONS

This paper proposed Polar-ToyBar, a novel dual-polarized fast group-sparse phase retrieval algorithm for robust DOA estimation. The problem is formulated as a joint group-sparse model that couples information from horizontal and vertical channels. This non-convex problem is efficiently solved by FISTA after transforming it into a convex surrogate via the Majorization-Minimization technique. Furthermore, magnitude-only spatial ambiguities are avoided by employing a specifically designed dual-subarray structure with a quarter-wavelength spacing ($d = \lambda/4$). Simulations demonstrate that Polar-ToyBar achieves superior estimation accuracy and robustness compared to standard ToyBar, while simultaneously enabling the recovery of polarization information.

REFERENCES

- [1] F. Wen, J. Shi, and Z. Zhang, "Direction finding for bistatic MIMO radar with unknown spatially colored noise," *Circuits Syst. Signal Process.*, vol. 39, no. 5, pp. 2412–2424, May 2020.
- [2] H. L. Van Trees, *Optimum Array Processing: Part IV of Detection, Estimation, and Modulation Theory*. New York, NY, USA: John Wiley & Sons, 2002.
- [3] R. O. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Trans. Antennas Propag.*, vol. 34, no. 3, pp. 276–280, Mar. 1986.
- [4] R. Roy and T. Kailath, "ESPRIT-Estimation of signal parameters via rotation invariance techniques," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 37, no. 7, pp. 984–995, Jul. 1989.
- [5] R. W. Gerchberg and W. O. Saxton, "A practical algorithm for the determination of phase from image and diffraction plane pictures," *Optik*, vol. 35, no. 2, pp. 237–246, 1972.
- [6] E. J. Candès, T. Strohmer, and V. Voroninski, "PhaseLift: Exact and stable signal recovery from magnitude measurements via convex programming," *Commun. Pure Appl. Math.*, vol. 66, no. 8, pp. 1241–1274, Aug. 2013.
- [7] E. J. Candès, X. Li, and M. Soltanolkotabi, "Phase retrieval via Wirtinger flow: Theory and algorithms," *IEEE Trans. Inf. Theory*, vol. 61, no. 4, pp. 1985–2007, Apr. 2015.
- [8] Y. Shechtman, A. Beck, and Y. C. Eldar, "GESPAR: Efficient phase retrieval of sparse signals," *IEEE Trans. Signal Process.*, vol. 62, no. 4, pp. 928–938, Feb. 2014.
- [9] H. Kim, A. M. Haimovich, and Y. C. Eldar, "Non-coherent direction of arrival estimation from magnitude only measurements," *IEEE Signal Process. Lett.*, vol. 22, no. 7, pp. 925–929, Jul. 2015.
- [10] Z. Wan and W. Liu, "Non-coherent DOA estimation based on a dual-array structure without reference signals," in *Proc. IEEE Int. Workshop Comput. Adv. Multi-Sensor Adapt. Process. (CAMSAP)*, Guadeloupe, West Indies, Dec. 2019, pp. 1–5.
- [11] Z. Wan and W. Liu, "A fast group sparsity based phase retrieval algorithm for non-coherent DOA estimation," in *Proc. 54th Asilomar Conf. Signals, Syst., Comput.*, Pacific Grove, CA, USA, Nov. 2020, pp. 315–319.
- [12] Z. Wan and W. Liu, "Non-coherent DOA estimation via proximal gradient based on a dual-array structure," *IEEE Access*, vol. 9, pp. 26792–26801, 2021.
- [13] T. Qiu, P. Babu, and D. P. Palomar, "PRIME: Phase retrieval via majorization-minimization," *IEEE Trans. Signal Process.*, vol. 64, no. 19, pp. 5174–5186, Oct. 2016.
- [14] A. Beck and M. Teboulle, "A fast iterative shrinkage-thresholding algorithm for linear inverse problems," *SIAM J. Imaging Sci.*, vol. 2, no. 1, pp. 183–202, Mar. 2009.
- [15] N. Parikh and S. Boyd, "Proximal algorithms," *Found. Trends Optim.*, vol. 1, no. 3, pp. 123–231, Jan. 2014.
- [16] A. Beck and M. Teboulle, "Gradient-based algorithms with applications to signal recovery problems," in *Convex Optimization in Signal Processing and Communications*, D. P. Palomar and Y. C. Eldar, Eds. Cambridge, U.K.: Cambridge Univ. Press, 2010, pp. 42–88.
- [17] Y. E. Nesterov, "A method for solving the convex programming problem with convergence rate $O(1/k^2)$," *Soviet Math. Dokl.*, vol. 27, no. 2, pp. 372–376, 1983.